Julius Alexander Zeiss, M.Sc. Physics

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Profile

- I develop classical algorithms that make difficult quantum problems tractable. Combining group theory, quantum Shannon theory, and convex optimization, I use quantum de Finetti theorems and symmetry-reduced semidefinite programming hierarchies to obtain rigorous bounds on quantum correlations and quantum error correction. Alongside these algorithmic questions, I study the power and limitations of entanglement in communication tasks and non-local games, especially quantum graph coloring. My work connects quantum information with graph theory and combinatorics, investigating how entanglement changes classical constraints and how graph structure shapes quantum strategies.
- Speaker at leading international conferences (TQC, Beyond IID, AQIS) and invited speaker at institutions including Stanford University and the Perimeter Institute for Theoretical Physics.
- Founder and host of RWTH Aachen's regular A.I. Workshop on rigorous, reproducible science with A.I.

Education

since 03/2023	**PhD in Physics (Quantum Information Theory), RWTH Aachen University, Germany** Institute for Quantum Information, final-year PhD student Expected completion: 31 March 2027 Advisor: Prof. Dr. Mario Berta
10/2019 – 08/2022	**M.Sc. Physics, University of Cologne & University of Bonn, Germany** Focus: theoretical and mathematical physics Thesis: *SDP Relaxations for Quantum Inflation* Advisor: Prof. Dr. David Gross, Thesis grade: 1.0 (very good) Overall grade: 1.4 (very good)
10/2018 – 08/2019	**Master's studies in Physics, University of Münster, Germany** Transferred to Cologne/Bonn Focus: theoretical physics, quantum information theory, biophysics
10/2015 – 09/2018	**B.Sc. Physics, University of Münster, Germany** Thesis: *Numerical Treatment of the Manning Potential for the Description of Ammonia* Advisor: Prof. Dr. Gernot Münster, Thesis grade: 1.0 (very good) Overall grade: 1.8 (good)
04/2013 – 09/2015	**Bachelor's studies in Business Administration, University of Münster, Germany** Transferred to physics
09/2010 – 05/2012	**International Baccalaureate, King William's College, Isle of Man** Overall grade: 41/45 points (very good, 1.1 German Abitur)

Publications and Preprints

2026	**J. A. Zeiss**, *Comment on "The polarization hierarchy for polynomial optimization over convex bodies, with applications to nonnegative matrix rank"*, arXiv:2610.24542 (submitted to Linear Algebra and its Applications)
2026	**J. A. Zeiss**, *A counterexample to the quantum Hedetniemi conjecture*, arXiv:2609.20690
2026	**J. A. Zeiss**, *Optimal entanglement-assisted source coding under a balanced-difference promise*, arXiv:2609.15757 • Lean formalization
2026	M. Berta, P. Costa Rico, G. Koßmann, L. Lami, **J. A. Zeiss**, *Sharp continuity of quantum conditional entropy*, arXiv:2607.24687 (under review at Annales Henri Poincaré)

2026 G. Koßmann, **J. A. Zeiss**, *Fixed points in de Finetti hierarchies*, arXiv:2607.23689 (under review at Communications in Mathematical Physics)

2026 **J. A. Zeiss**, G. Koßmann, R. Schwonnek, M. Plávala, *Finite de Finetti for convex bodies and polynomial optimization*, arXiv:2601.15184 (under review at Mathematical Programming A)

2025 **J. A. Zeiss**, G. Koßmann, O. Fawzi, M. Berta, *Approximating fixed size quantum correlations in polynomial time*, arXiv:2507.12302 (under review at PRX Quantum)

2025 G. Koßmann, **J. A. Zeiss**, O. Fawzi, M. Berta, *On approximate quantum error correction for symmetric noise*, arXiv:2507.12326 (under review at Quantum)

Invited Talks

06/2026 **Stanford University, CS Theory Group, Palo Alto, USA**
A Family of Information-theoretic de Finetti Theorems for Constrained Optimization

04/2026 **University of Cologne, Quantum Information Seminar, Germany**
Approximating fixed size quantum correlations in polynomial time

07/2023 **Siegen University, Germany**
Inner approximation schemes for approximate quantum error correction

02/2023 **Perimeter Institute for Theoretical Physics, Waterloo, Canada**
Infinite Dimensional Optimisation Problems in Quantum Information — An operator algebra approach to the NPA Hierarchy

05/2022 **Alpen-Adria-Universität Klagenfurt, Austria**
SDP methods for the quantum causal compatibility problem

Contributed Talks

09/2026 **TQC 2026, Sherbrooke, Canada**
A Family of Information-Theoretic de Finetti Theorems for Constrained Optimization

05/2026 **ILAS 2026, Virginia Tech, Blacksburg, USA**
Convergent Inner-Outer Approximation Schemes From De Finetti Theorems For Games And Quantum Error Correction

05/2026 **ILAS 2026, Virginia Tech, Blacksburg, USA**
Information-theoretic finite de Finetti theorems for quantum and beyond[*]

03/2026 **Quantum at the Dunes, IIP-UFRN, Natal, Brazil**
Approximating fixed size quantum correlations in polynomial time

08/2025 **AQIS 2025, The University of Hong Kong, Hong Kong**
Approximating fixed size quantum correlations in polynomial time

07/2025 **13th Beyond IID in Information Theory, TUM Institute for Advanced Study, Garching, Germany**
Approximating fixed size quantum correlations in polynomial time

07/2025 **13th Beyond IID in Information Theory, TUM Institute for Advanced Study, Garching, Germany**
On approximate quantum error correction for symmetric noise[*]

[*] Talk given by a co-author.

Posters

04/2026 **851. WE-Heraeus-Seminar: 90 Years of Einstein-Podolsky-Rosen Paradox, Bad Honnef, Germany**
Approximating fixed size quantum correlations in polynomial time

01/2026 **QIP 2026, Riga, Latvia**
Approximating fixed size quantum correlations in polynomial time

08/2025 **QMATH Masterclass on Representation Theory in Quantum Information Science, University of Copenhagen, Denmark**
Schur–Weyl Duality for Efficient Approximation of Constrained Separability

07/2024 **Beyond IID in Information Theory, University of Illinois Urbana-Champaign, USA**
Approximating fixed size quantum correlations in polynomial time

Teaching and Supervision

since 03/2023 — **RWTH Aachen University, Germany**
Head tutor and organizer: Advanced Seminar on Bell Inequalities (winter semester 2026/27)
Supervised two master's and two bachelor's theses in quantum information theory
Tutorials: Quantum Information Theory (master's; 2024, 2025), Quantum Mechanics (bachelor's; 2025), Statistical Physics (bachelor's; 2023, 2024)
Master's seminar: Advanced Topics in Quantum Information Theory (2024, 2026)

04/2021 – 04/2022 — **University of Cologne, Germany**
Tutorials: Linear Algebra & Analysis, Proofs in Mathematics, Theoretical Physics

04/2019 – 09/2019 — **University of Münster, Germany**
Tutorials: Applied Physics

Academic Service

Conference reviewing — TQC (2023, 2024, 2025) • ISIT (2023, 2024, 2025) • QIP (2024, 2025, 2026, 2027) • Beyond IID (2023, 2024)

Journal refereeing — IEEE Transactions on Information Theory (2025)

Outreach

since 2026 — **A.I. Workshop, RWTH Aachen University, Germany**
Founder and host of a regular workshop on A.I. tools for research (from coding assistants and version control to formally verified proofs) and on using them to conduct better science

2024 – 2026 — **ML4Q — Matter and Light for Quantum Computing, Germany**
Instructor: school internships (July 2024), Quanteen Day 2024, Quanteen Day 2026

11/2022 — **DPG Physik Journal**
A. Matthies and J. Zeiss, *Quantum Computing*, review article on the Quantum Computing Bad Honnef Physics School, Physik Journal 21 (2022) No. 11, p. 57

Skills

Languages — German (native), English (bilingual)

Coding — Python, Mathematica, Lean (formal verification), Git, GitHub, Fortran, PHP, JavaScript, HTML, SQL, VBA

Other Professional Experience

2014, 2015 — **Fielmann AG, Hamburg, Germany**
DCF modeling with Monte Carlo simulations, VBA and SQL solutions

01/2013 – 02/2013 — **HMT, Hamburg, Germany**
Mathematical modeling using Monte Carlo simulations

08/2011 — **AKO Capital, London, England**
Internship in financial research using mathematical modeling